

1. Arithmetic Operators

(+, -, *, /, %, //, **)

a = 15

b = 4

print(a + b) # 19

print(a - b) # 11

print(a * b) # 60

print(a / b) # 3.75

print(a % b) # 3 (remainder)

print(a // b) # 3 (floor division)

print(a ** b) # 50625 (15^4)

2. Comparison Operators

(==, !=, >, <, >=, <=)

x = 10

y = 20

print(x == y) # False

print(x != y) # True

print(x > y) # False

print(x < y) # True

print(x >= 10) # True

print(y <= 15) # False

3. Logical Operators

(and, or, not)

a = 5

b = 8

print(a > 3 and b > 5) # True

print(a > 10 or b > 5) # True

print(not(a > 3)) # False

4. Assignment Operators

(=, +=, -=, *=, /=)

```
x = 10
```

```
x += 5 # x = x + 5 → 15
```

```
x *= 2 # x = x * 2 → 30
```

```
x -= 10 # x = x - 10 → 20
```

```
print(x)
```

5. Membership Operators

(in, not in)

```
name = "Sanjana"
```

```
print("a" in name) # True
```

```
print("z" not in name) # True
```

6. Identity Operators

(is, is not)

```
a = [1, 2]
```

```
b = [1, 2]
```

```
print(a == b) # True (values same)
```

```
print(a is b) # False (different memory)
```